Seeed Studio Tiny Frame Interface

Breathing and Heartbeat

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1. Application project

For different application projects, all messages related to TF frames are listed for users to refer to and complete the analysis. The message classes and message data bits that appear in the document are equipped with corresponding actual projects.

1.1. Breathing + Heart Rate Detection Project

Message type: Report phase test result 0 x 0A13

The message type is 0x0A13 and only supports one-way data transmission mode .

The radar sends data to the host computer					
Format	Number of bytes	basic type	frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	start frame	01	Used to output the total phase, heartbeat phase, and respiratory phase results.
ID	2 bytes	uint16	Frame ID	00 00	
LEN	2 bytes	uint16	Data frame length	00 04	
TYPE	2 bytes	uint16	Frame type	0A 13	
HEAD_CKSUM	1 byte	uint8	header checksum	\	
DATA	4 bytes	float	[total phase]	\	
DATA	4 bytes	float	[breath phase]	\	
DATA	4 bytes	float	[heart phase]	\	
DATA_CKSUM	1 byte	uint8	Data checksum	\	

Message type: Reporting respiratory rate test results 0x0A14

The message type is 0x0A14 and only supports one-way data transmission mode .

The radar sends data to the host computer					
Format	Number of bytes	basic type	frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	start frame	01	Used to report respiratory rate test results.
ID	2 bytes	uint16	Frame ID	00 00	
LEN	2 bytes	uint16	Data frame length	00 04	
TYPE	2 bytes	uint16	Frame Type	0A 14	
HEAD_CKSUM	1 byte	uint8	Header Checksum	\	
DATA	4 bytes	float	[rate]	\	
DATA_CKSUM	1 byte	uint8	Data checksum	\	

Message type: Report heart rate test results 0x0A15

The message type is 0x0A15 and only supports one-way data transmission mode .

The radar sends data to the host computer					
Format	Number of bytes	basic type	frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	start frame	01	Used to report heartbeat phase test results.
ID	2 bytes	uint16	Frame ID	00 00	
LEN	2 bytes	uint16	Data frame length	00 04	
TYPE	2 bytes	uint16	Frame type	0A 15	
HEAD_CKSUM	1 byte	uint8	header checksum	\	
DATA	4 bytes	float	[rate]	\	
DATA_CKSUM	1 byte	uint8	Data checksum	\	

Message type: Report detection target distance 0x0A16

The message type is 0x0A16 and only supports one-way data transmission mode .

The radar sends data to the host computer					
Format	Number of bytes	basic type	frame structure	Example frame	Frame meaning
SOF	1 byte	uint8	start frame	01	Used to report detection distance.
ID	2 bytes	uint16	Frame ID	00 00	
LEN	2 bytes	uint16	Data frame length	00 04	
TYPE	2 bytes	uint16	Frame type	0A 16	
HEAD_CKSUM	1 byte	uint8	header checksum	\	
DATA	4 bytes	uint 32	[flag]	\	
DATA	4 bytes	float	[range]	\	
DATA_CKSUM	1 byte	uint8	Data checksum	\	

Note: When the flag is 1, the output distance (unit: cm)

When the flag is 0, no distance is output

A. The following is the data conversion of the D ATA bit:

Convert to float: For example, the [rate] bits are 0x66, 0x66, 0xA2, 0x41. First spell it into a uint 32 -bit integer. Since the TF frame Data bit is little endian, the value is 0x 41A26666, and then perform float type enhancement Turn, the final result is: 20.3 .

```
int main(void )
{
unsigned int param = 0x41A26666;
    float res = *( float *)&param;

printf( "data: %f\n" , res);
    return 0;
}
```

B. The following is the analysis of each CKSUM :

HEAD_CKSUM : TF frame header checksum [from the first byte to the previous byte of the HEAD_CKSUM bit]

DATA_CKSUM : TF data checksum [the first byte of D ATA to the previous byte of DATA_CKSUM bit]

The method c code for calculating C KSUM is as follows:

```
unsigned char getCksum(unsigned char *data, unsigned char len)
{
unsigned char ret = 0;

   for ( int i = 0; i < len; i++)
ret = ret ^ data[i];

ret = ~ret;

   return ret;
}

```

2. programming interface

2.1. Encode TF message

void tinyFramefTx(TF_TYPE type, uint8 *data, TF_LEN len) ;

Type is the sending data type, uint16 type. For example, when personnel detection data results are reported, the data type is 0x0A10.

Uint8* data is the address to send data.

Len is the length of the sent data, uint16 type.

2.2. Decode TF message

TinyFrameRx tinyFramefRx(void) ;

After successfully receiving the message, the received data is returned to a variable of type TinyFrameRx .